

App Dev – 2

Project Report

1. Student Details

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About Me: I am a student at the IIT Madras BS Degree program with a deep interest in full-stack web application development. I enjoy building practical, scalable applications that bridge backend logic with clean frontend experiences. This project gave me hands-on exposure to Vue.js and asynchronous job processing, both of which were new to me during development.

2. Project Details

Project Title: Hospital Management Systems

Problem Statement:

To design and build a multi-role, full-stack web application for managing hospital operations. The system supports three roles — Admin, Doctor, and Patient — each with tailored dashboards and functionality.

Approach:

The application was built using Flask as the backend API server and Vue.js (via CDN) as the single-page frontend framework, communicating through RESTful JSON APIs. The frontend uses Vue Router for client-side navigation without page reloads.

Patients can:

- Register, login, and edit their profile.
- Search doctors and departments.
- Check doctor availability and book or cancel appointments.
- View their complete treatment history and prescription details.

Doctors can:

- Login and view today's and upcoming appointments on their dashboard.

- Provide availability slots for the next 7 days, with select-all/clear functionality.
- Attend patients by recording diagnosis and prescriptions (per meal timing and quantity).
- Cancel appointments and view any patient's full treatment history.

Admin can:

- Manage doctors — add, edit, delete, blacklist/allowlist.
- Manage patients — edit, delete, blacklist/allowlist.
- Manage departments — create, edit, delete (with guard against deleting occupied departments).
- View and monitor all appointments — today's, upcoming, and previous.
- Search across doctors, patients, and departments in real time from any page.
- Receive system-level visibility across the entire hospital.

3. AI/LLM Declaration

I used Claude (Anthropic) and Chatgpt to:

- Assist in the logic of creating check availability, book appointments, providing treatments and for CSV report
- Little Assistance in debugging.

Extent of usage: ~15–20%, limited to specific integration patterns and debugging. All component logic, API design, database modeling, and UI structure were designed and implemented manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework and API server
SQLAlchemy	ORM for SQLite database
SQLite	Lightweight embedded relational database
Vue.js 3 (CDN)	Reactive single-page frontend framework
Vue Router 4 (CDN)	Client-side routing between dashboard views
Axios	HTTP client for Vue ↔ Flask API communication
Google Fonts (Lora)	Typography styling

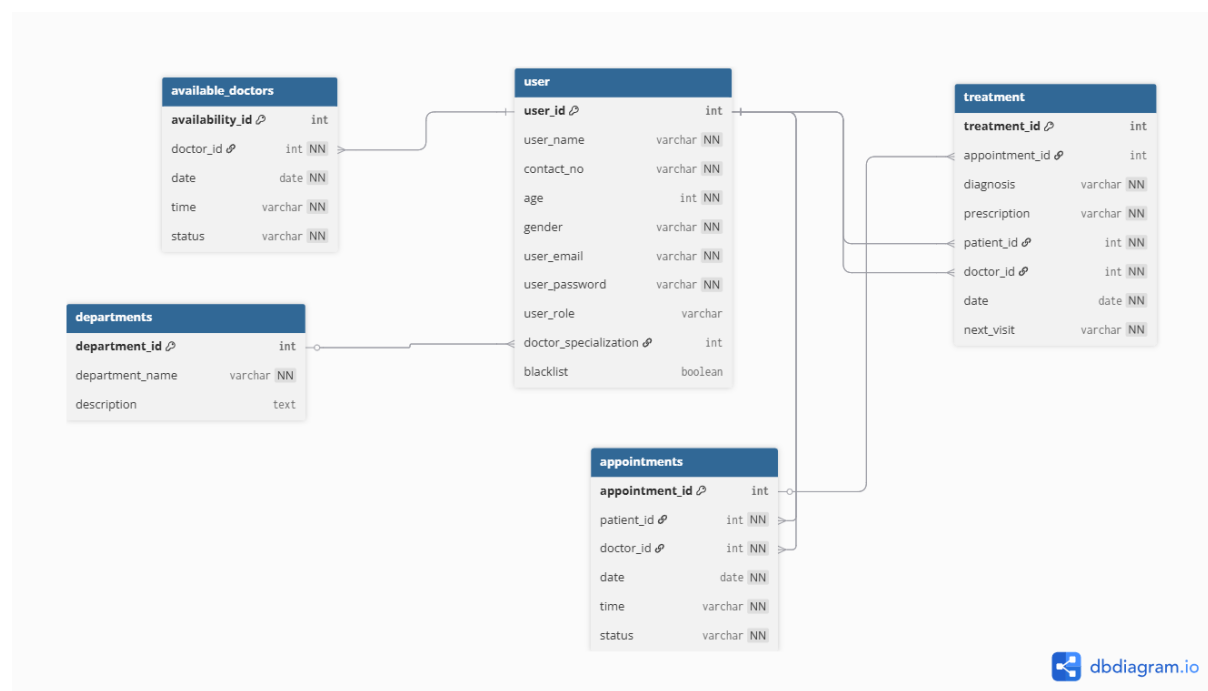
5. Database Schema / ER Diagram

Tables:

- **User** — stores all users across roles (*user_id*, *user_name*, *contact_no*, *age*, *gender*, *user_email*, *user_password*, *user_role*, *doctor_specialization* [FK → *Department*], *blacklist*)
- **Department** — stores hospital departments (*department_id*, *department_name*, *description*)
- **Appointments** — stores appointment bookings (*appointment_id*, *patient_id* [FK → *User*], *doctor_id* [FK → *User*], *date*, *time*, *status*)
- **Treatment** — stores treatment records linked to appointments (*treatment_id*, *appointment_id* [FK → *Appointments*], *diagnosis*, *prescription* [JSON], *patient_id* [FK → *User*], *doctor_id* [FK → *User*], *date*, *next_visit*)
- **Availability** (available_doctors) — stores doctor-provided time slots (*availability_id*, *doctor_id* [FK → *User*], *date*, *time*, *status*)

Relationships:

- One-to-Many → Department → User (one department can have many doctors)
- One-to-Many → User (patient) → Appointments (one patient can have many appointments)
- One-to-Many → User (doctor) → Appointments (one doctor can have many appointments)
- One-to-Many → Appointments → Treatment (each appointment can produce one treatment record)
- One-to-Many → User (doctor) → Availability (one doctor can have many available slots)



6. Architecture and Features

Architecture Overview:

app.py → Flask application, all routes and models
/templates/index.html → Single HTML file containing all Vue components
and Vue Router configuration
/static/ → Images (logos, icons)
database.sqlite3 → SQLite database (auto-created on first run)

frontend is a true single-page application — all Vue components (`index`, `login`, `signup`, `admin_dashboard`, `doctor_dashboard`, `patient_dashboard`, and ~25 others) live in a single `index.html` file and are rendered by Vue Router using hash-based history. Flask serves only the initial HTML and all subsequent communication is via JSON APIs prefixed with `/api/`.

Implemented Features:

Authentication & Sessions

- Signup with validation (email uniqueness, age, contact number format, password confirmation)
- Login with role-based redirect (admin / doctor / patient)
- Server-side session management via Flask sessions
- Logout with session clear

Admin Dashboard

- Real-time search across doctors, patients, and departments from any admin page
- Today's appointments overview on the home dashboard
- Add, edit, delete, blacklist/allowlist doctors and patients
- Department management with guard (cannot delete if doctors are assigned)
- Appointment history split into upcoming and previous views
- Blacklist management as a dedicated page

Doctor Dashboard

- Today's and upcoming appointment views
- Attend patient flow — diagnosis input, multi-medicine prescription with per-meal checkboxes and quantity selectors
- Availability provision — 7-day calendar with per-slot toggle and select-all/clear per day
- Patient history viewer — lookup any patient's past treatment records

- Summary stats — patients treated, slots provided, missed/cancelled appointment counts

Patient Dashboard

- Book appointments by browsing departments → doctors → available slots
- Appointment booking with conflict checks (same department, same doctor, already booked slot)
- Cancel appointment
- View full treatment history with prescription breakdown (medicine, meal timing, quantity)
- Edit personal profile

7. Video Presentation

https://drive.google.com/file/d/1lvjNeP-drd_YI5fKDm3dEiOfLvG5wO9j/view?usp=sharing